

1 Introduction

The volatility of a basket of random variables is given by

$$\sigma_{\text{basket}}^2 = \sum_i^N w_i \sigma_i^2 + \sum_i^N \sum_{j \neq i}^N w_i w_j \sigma_i \sigma_j \rho_{ij}$$

If the correlation of the components of a basket increases, then the volatility will too. The price of an option is directly related to its volatility. In particular, option price increases with volatility (vega > 0). This is also the economic basis of the trade: an index option prices in the correlation between its components, so index implied volatility tends to trade richer than the volatility implied by the individual names. That gap is the dispersion premium.

2 Assumptions

Our model will be based on the exposition presented in [1].

1. Correlation is mean-reverting so if it is low, it will **likely** go up, and if it high it will **likely** go down.
2. Lower average correlation implies lower (basket) index variance, which in turn implies that an option on the basket gets cheaper.

Note: In the referenced document the author makes the following inaccurate claim: The volatility of the sum of the components of a basket are equal to the overall basket's volatility (i.e. $\sigma_{\text{basket}} = \sum_i w_i \sigma_i$). Therefore the value of a basket of options should be equal to an option on the entire basket. The reason this is incorrect is because variance carries correlation cross-terms. In general, given that $\rho_{i,j} \leq 1$, we have that $\sigma_{\text{basket}} \leq \sum_i w_i \sigma_i$ with equality iff all pairwise correlations equal one. For $\rho < 1$ the index is less volatile than the collection of its members so an option on the basket is worth less than the basket of options (that gap is the dispersion premium). The relationship can still be captured by a linear approximation $\sigma_{\text{basket}} \cong \sqrt{\bar{\rho}} \sum w_i \sigma_i$ (where $\bar{\rho} = \frac{\sum_{i \neq j} w_i w_j \sigma_i \sigma_j \rho_{i,j}}{\sum_{i \neq j} w_i w_j \sigma_i \sigma_j}$). This follows from dropping the second term in the following identity $\sigma_{\text{basket}}^2 = \bar{\rho} (\sum w_i \sigma_i)^2 + (1 - \bar{\rho}) \sum w_i^2 \sigma_i^2$.

3 Strategy:

Our strategy will be based on the following idea:

1. If we expect correlation to rise $\uparrow$, then:
 - (a) Buy index.
 - (b) Sell singles (short dispersion).

2. If we expect correlation to fall ↓, then:

- (a) Sell index.
- (b) Buy singles (long dispersion).

The reason we sell singles is to **Hedge the Position**.

We also note the following statement in the document: ... *So if the average market correlation is unusually low, then the price of a basket of options will likely decrease as the price of an option on the basket increases.*

How do we know that volatility is low or high? We simply look at the volatility series and determine the historical percentile its in.

4 Execution:

We begin by establishing a **reference distribution**. We pick *SPY top 10* and compute daily returns.

1. Compute a rolling window average pairwise correlation among components.
2. Convert today's value to a z-score. **Unusually low** could mean $z < -1$ (bottom 16%).
3. Or implied correlation baseline computed from options.

4.1 Data Sources:

- **Prices:** Daily adjusted closes from Yahoo Finance via `yfinance` (`auto_adjust = True`).
- **Returns:** Business day log returns from those adjusted prices.

4.2 Signal Validity Results:

Does the z-score actually predict the direction of future correlation ?

Basket	Sampling	Regime	Signals	Hit Rate	Mean $\Delta\rho$
XLF	Monthly	Overall	25	88.0%	−0.015
	Monthly	HIGH	13	92.3%	−
	Monthly	LOW	12	83.3%	−
	Weekly	Overall	144	83.3%	+0.039
XLE	Monthly	Overall	28	85.7%	+0.036
	Monthly	HIGH	11	90.9%	−
	Monthly	LOW	17	82.4%	−
	Weekly	Overall	151	78.8%	+0.030

Table 1: A signal is “correct” if correlation rose after a **LOW** signal ($z < -1$) or fell after a **HIGH** signal ($z > +1$). Monthly sampling uses non-overlapping windows; weekly sampling overlaps and is shown for completeness. The edge is symmetric across regimes and replicates across both sectors.

4.3 P & L Backtest Results:

Metric	XLF (Financials)	XLE (Energy)
Period	2018-06 – 2026-06	
Initial capital	\$100,000	\$100,000
Final value	\$88,718	\$81,103
Total return	−11.3%	−18.9%
Benchmark (buy & hold)	+136.2%	+108.9%
Annualized return	−1.5%	−2.6%
Annualized volatility	2.9%	3.7%
Sharpe ratio	−0.52	−0.70
Max drawdown	−12.2%	−21.8%
Trade legs	407	341
Win rate (legs)	53.8%	59.5%
Total commission	\$11,387	\$9,371
LOW-regime P&L (gross)	−\$3,030	−\$5,902
HIGH-regime P&L (gross)	+\$2,979	−\$3,624

Table 2: Backtest performance, 2018–2026. Both sectors lose money despite the statistically valid signal. Annualized volatility is low ($\sim 3\%$) because the long-index / short-component structure is near market-neutral. On XLF the gross regime P&L nearly cancels (−\$3,030 vs +\$2,979), so the \$11,387 in commissions accounts for essentially the entire loss.

We make the following observations:

1. Both strategies lose money over 2018 – 2026 and significantly under-perform a buy

and hold position in the ETF. This comparison is illustrative rather than fair, since the strategy is near market-neutral and was never intended to track a long-only benchmark.

2. The strategy is approximately break even before costs and loses approximately its transaction costs.
3. The 2.9% annualized volatility shows how thoroughly the long-ETF/short-components construction neutralizes market risk. This neutrality is the design goal, but it also means the strategy captures only the thin realized-correlation drift, which is too small to survive trading frictions.

4.4 Limitations:

It is important to note that the strategy trades the wrong instrument. The dispersion premium lives in options but in this write-up I trade just the stocks and the ETF. Therefore I only capture a small part of the correlation move which isn't enough to cover trading costs. This is a reason that although we have a signal that predicts correlation well, it still loses money when traded.

4.4.1 Modeling Costs:

Flat 10 basis points per trade with no modeling of slippage, market impact or borrow cost on short positions. Fills are also taken at the signal-day close rather than the next open, which is mildly optimistic.

4.4.2 Validity Test:

We should note that the validity test rests on a small number of independent monthly observations (roughly 25–28 per sector), so although the hit rates are encouraging, they are not statistically airtight.

We finally note that the constituent lists are held fixed at their current membership over the whole period, so the real index membership may have changed over time, and the study covers only two sectors during what was largely a bull market.

References

- [1] Max Dama. *Max Dama on Automated Trading*. 2011. URL: <http://isomorphisms.sdf.org/maxdama.pdf>.